MEMBRANE DISC DIFFUSER

DN270

Fine bubble · 10 inch

1 – 6 m³/h
FLOW RATE

1 – 3 mm
BUBBLE SIZE

6500
PORES

3/4" NPT
CONNECTOR

APPLICATION

- Aeration
- SBR reaction basin
- Activated sludge process
- Wastewater treatment process
- Denitrification aerobic process
- Aquafarm and other aeration applications

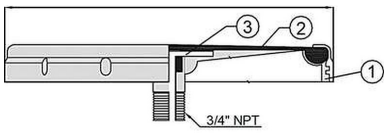
MEMBRANE DISC DIFFUSER

TECHNICAL DATA | FINE BUBBLE DISC DIFFUSER

Redpump
Italy Standard



CONSTRUCTION



1	Retaining ring	PP
2	Membrane	EPDM
3	Disc	PP

PP: polypropylene. EPDM: ethylene propylene diene monomer.

FEATURES

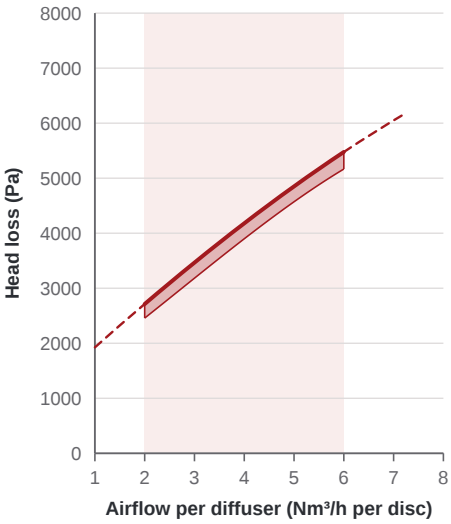
- High oxygen transfer efficiency (OTE): a large number of identical slits (pores) on the EPDM membrane produce fine air bubbles of 1 – 3 mm on average, saving power and lowering operating cost.
- Excellent backflow prevention with check valve, which avoids clogging and keeps sewage water from entering the aeration pipeline when the air supply is shut off (also available without check valve).
- EPDM membrane with high tensile strength, tear resistance and non-deformability. Its surface carries over 6500 micro pores. In operation the bubbles rise in a spiral whirlpool, giving the membrane a self-cleaning ability and a long service life.
- Unique seal ring provides tight sealing between the diffuser base and the membrane washer.
- PVC soft connector for installation on the pipeline is available as an optional accessory.

SPECIFICATION

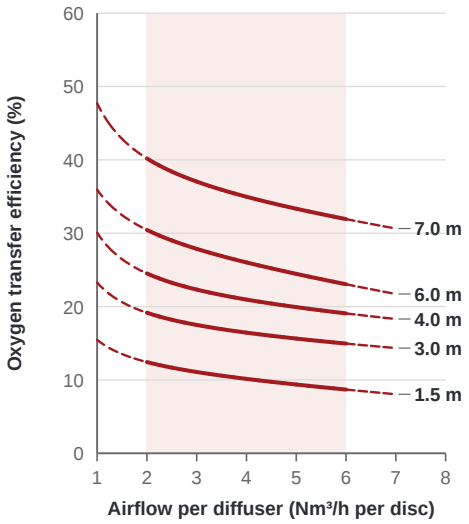
Disc diameter	**270 mm (10 inch)**	Service area	**0.5 m²**
Flow rate	**1 – 6 m³/h**	Pores quantity	**6500**
Design flow	**2.0 – 6.0 m³/h**	Bubble size	**1 – 3 mm**
Pressure loss	**2000 – 4800 Pa**	Max. temperature	**100 °C**
Range depth	**1.5 – 7.0 m**	Connector thread	**3/4" NPT**

PERFORMANCE CURVE

HEAD LOSS CURVE



OTE CURVE BY WATER DEPTH



Shaded area: design flow 2.0 – 6.0 m³/h. Dashed line: outside the design flow.

MEMBRANE DISC DIFFUSER

DN128

Coarse bubble · 5 inch

6 – 24 m³/h
FLOW RATE

17 %
OTE @ 4 m DEPTH

1550 Pa
HEAD LOSS @ 15 m³/h

3/4" NPT
CONNECTOR

APPLICATION

- Aeration
- Wastewater treatment
- Activated sludge process
- Water purification process
- Mixing
- Equalization process

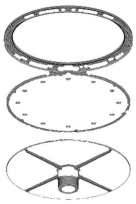
MEMBRANE DISC DIFFUSER

TECHNICAL DATA | COARSE BUBBLE DISC DIFFUSER

Redpump
Italy Standard



CONSTRUCTION



1	Rubber cap	EPDM
2	Main body	ABS

EPDM: ethylene propylene diene monomer. ABS: acrylonitrile butadiene styrene.

FEATURES

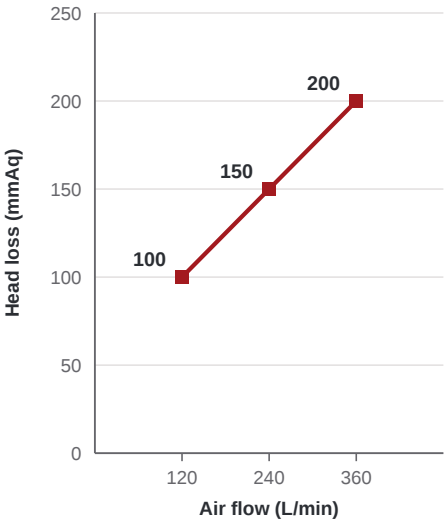
- Special structure prevents water backflow during air flow interruptions.
- Main body made of high-quality ABS with heat resistance; resists softening at high temperature.
- The membrane offers durable protection against corrosion, weather, oil and deformation.
- Adjustable discharge holes for different air volumes reduce pressure loss, prevent clogging and keep maintenance low.
- Easy installation; can be installed facing upwards or downwards.

SPECIFICATION

Disc diameter	**128 mm (5 inch)**	Flow rate	**6 – 24 m³/h**
Rubber cap	**EPDM**	OTE	**17 % @ water depth 4 m**
Main body	**ABS**	Head loss	**1550 Pa @ 15 m³/h**
Connector thread	**3/4" NPT (male)**		

PERFORMANCE CURVE

HEAD LOSS CURVE



OTE CURVE

